

Initialize the environment

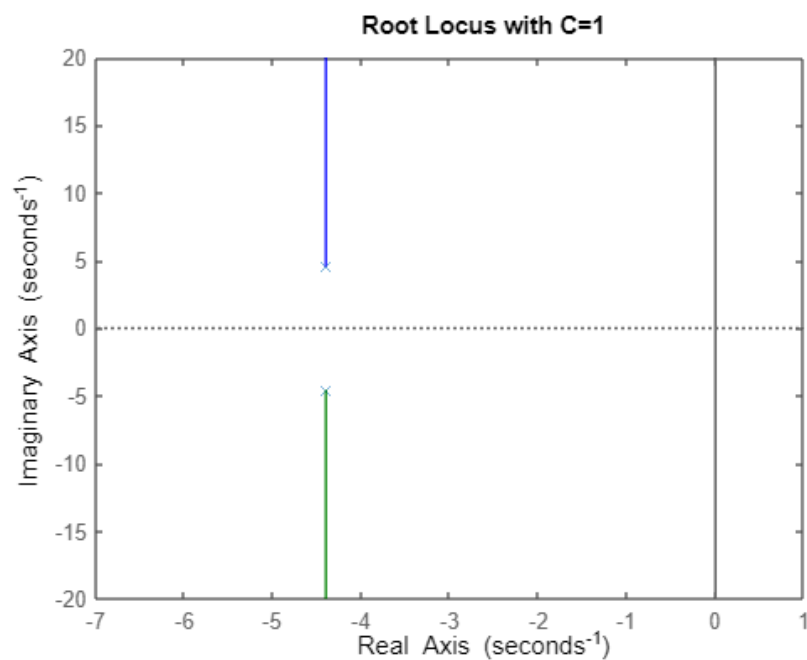
```
clear all;  
close all;
```

Define the Laplace variable and the system

```
s = tf('s');  
G = 1/(s^2 + 8.8*s + 40);
```

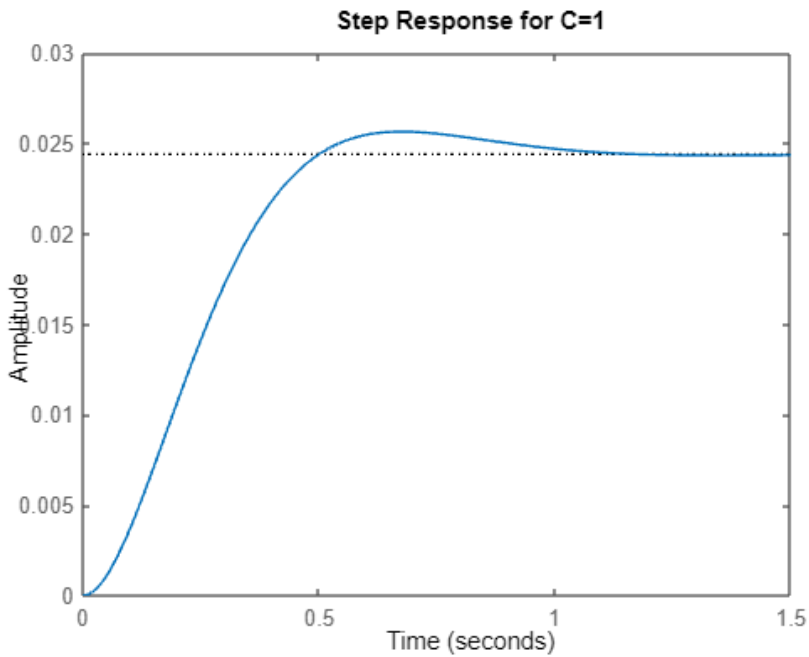
Root locus with C=1

```
C = 1;  
figure;  
rlocus(C*G);  
title('Root Locus with C=1');
```



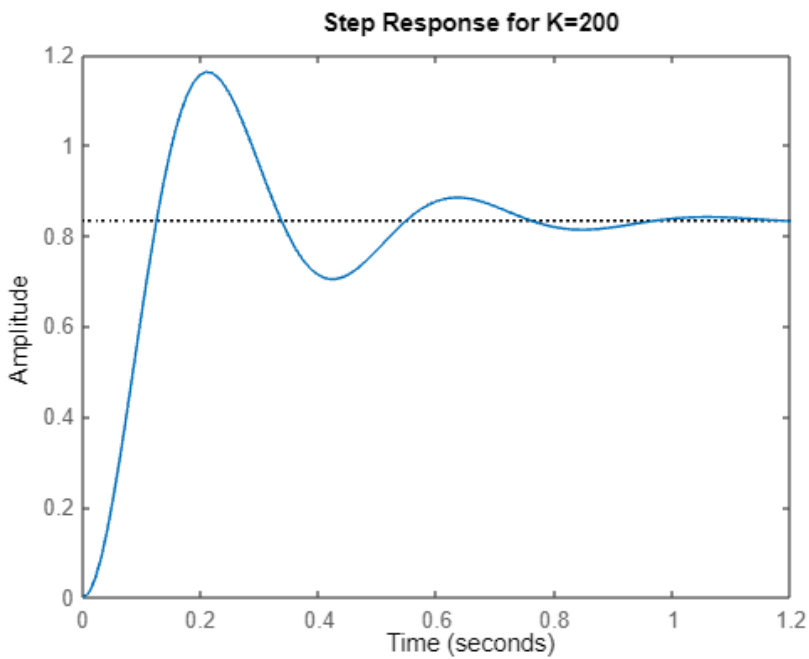
Step response for C=1

```
GCL = minreal(feedback(C*G, 1));  
figure;  
step(GCL);  
title('Step Response for C=1');
```



Increase gain for improved steady state response

```
K = 200;
C = K;
GCL = minreal(feedback(C*G, 1));
figure;
step(GCL);
title('Step Response for K=200');
```

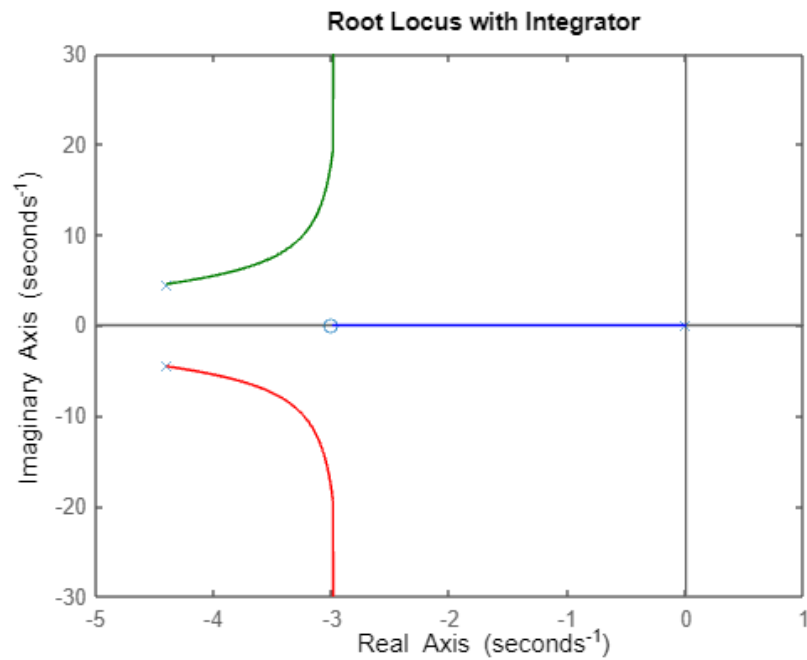


Root locus with an integrator

```

kp = 1;
ki = 3;
C = kp + ki/s;
figure;
rlocus(C*G);
title('Root Locus with Integrator');

```



Step response with K=50 and an integrator

```

K = 50;
C = K*(kp + ki/s);
GCL = minreal(feedback(C*G, 1));
figure;
step(GCL);
title('Step Response for K=50 with Integrator');

```

